

How Small is Small?

Non-linearities in Heterogeneous Agent Models

Javier Bianchi¹ Greg Kaplan²

¹Federal Reserve Bank of Minneapolis

²University of Chicago, e61 Institute & NBER

Overview

- Heterogeneous-agent (HA) models have become a workhorse framework for macroeconomic analysis
 - ▶ Capture salient features of household spending behavior
 - ▶ Match the level and distribution of MPCs in the data

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 - ▶ Well-understood that local methods are *accurate for small enough shocks*
- BUT: How small is small?
 - Our answer: for fiscal transfers, “small” must be *very small*.

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 - ▶ At the borrowing constraint, consumption is locally linear
 - ▶ MPC drops discontinuously once the constraint becomes slack
- Local methods are unreliable when aggregate responses are driven by failures of Ricardian equivalence
 - ▶ Shocks where they *are* accurate are those where $HA \approx RA$.

Roadmap

1. Partial equilibrium model and calibration
2. Fiscal stimulus in general equilibrium
3. Other shocks

Environment: Household Problem

$$\max_{\{c_{i,t}, a_{i,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta_i^t u(c_{i,t})$$

subject to

$$c_{i,t} + a_{i,t} \leq (1+r)a_{i,t-1} + (1-\tau)y_{i,t} + T_t \quad \text{and} \quad a_{i,t} \geq 0.$$

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Intertemporal Euler equation:

$$u'(c_{i,t}) \geq \beta_i(1+r)\mathbb{E}_t u'(c_{i,t+1})$$

with equality if $a_{i,t} > 0$.

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Aggregate consumption function:

$$C_t(\{T_s\}) = \int c_t(a, y, \beta) d\Gamma_t(a, y, \beta)$$

Fiscal Transfers

Starting from steady state, a one-time lump-sum transfer at $t = 0$:

$$T_0 = T + \Delta, \quad T_t = T \text{ for } t > 0.$$

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Non-linear solution

- Solve individual household problem exactly and aggregate

Linear solution

- Taylor expansion of aggregate consumption function around steady state

$$\mathcal{C}_t^{\text{lin}}(\Delta) = \mathcal{C}_t(0) + \left. \frac{\partial \mathcal{C}_t}{\partial \Delta} \right|_{\Delta=0} \cdot \Delta$$

Calibration

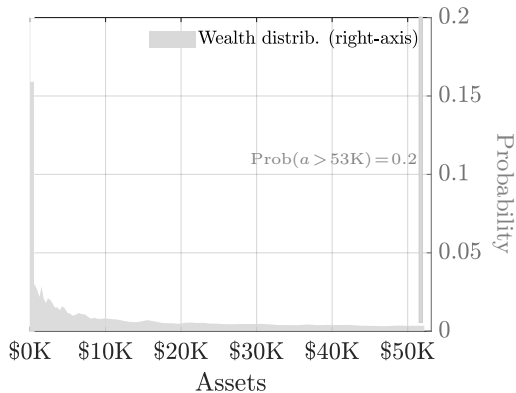
- A period = a quarter. Normalize average quarterly income to 1.
- Earnings process, $y_{i,t}$, subject to temporary and persistent shocks
 - ▶ Calibrated as in Kaplan-Violante (2022) to match distribution of households' earnings
- Interest rate 2% annual.
- Tax rate (τ) = 27.4%
- Steady-state transfers = 15% of GDP ($\approx$ \$25K/yr)

Calibrated to match moments

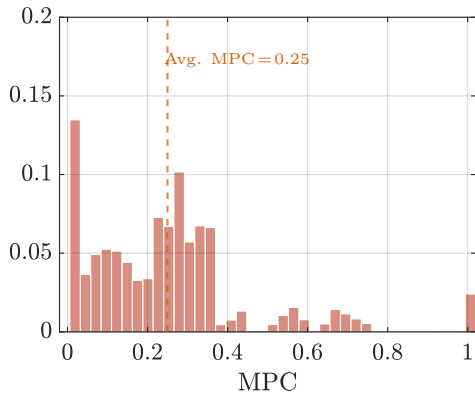
$$\left. \begin{array}{l} \text{Mean wealth / annual GDP} = 100\% (\approx \$166\text{K}) \\ \text{Avg. quarterly MPC (out of \$1,000)} = 25\% \end{array} \right\} \Rightarrow \bar{\beta} = 0.972, \sigma_{\beta} = 0.010$$

Steady-state wealth distribution and MPCs

(a) Wealth distribution

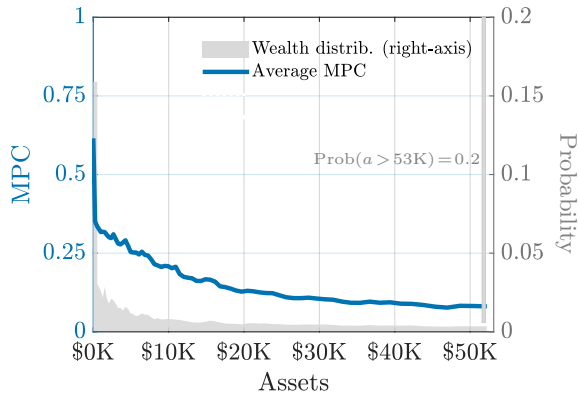


(b) MPC distribution

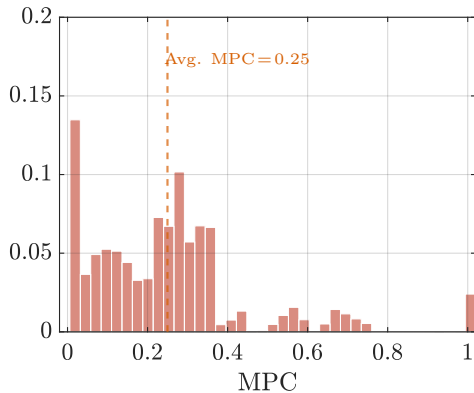


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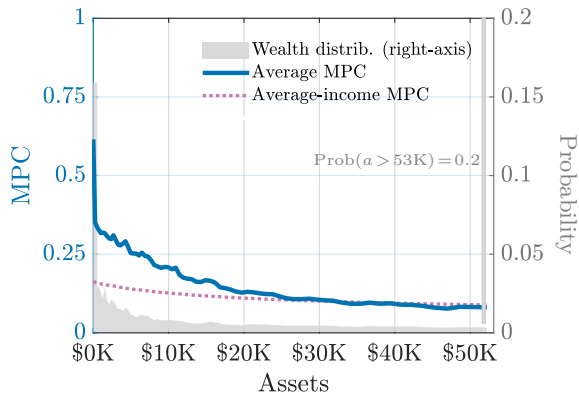


(b) MPC distribution

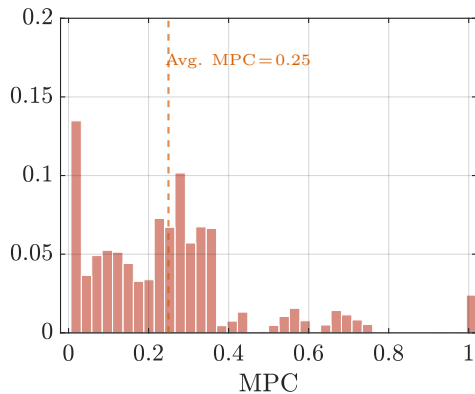


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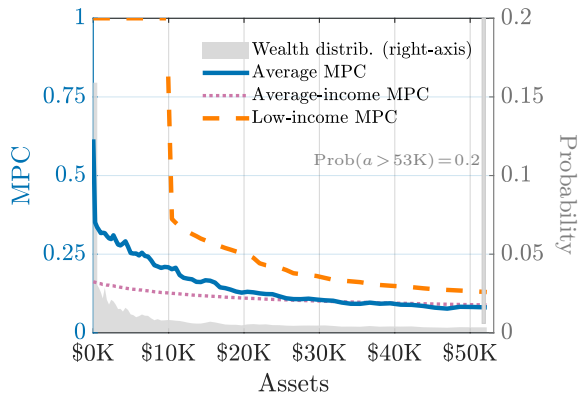


(b) MPC distribution

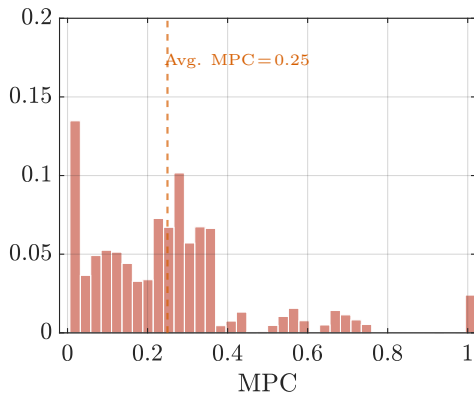


Steady-state wealth distribution and MPCs

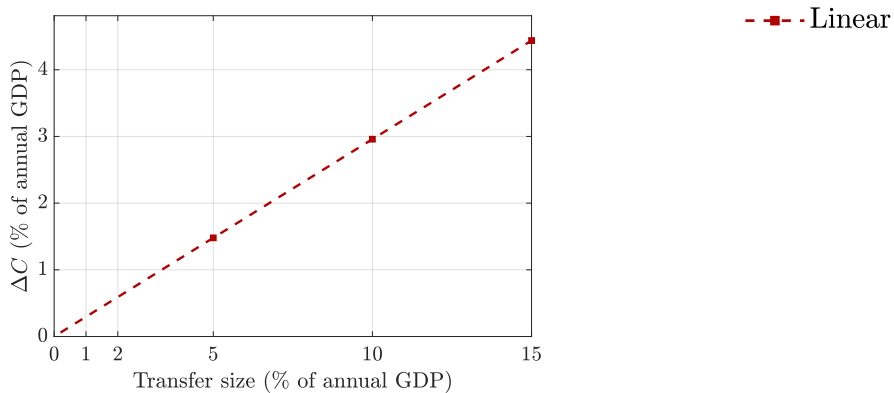
(a) Wealth distribution and MPCs



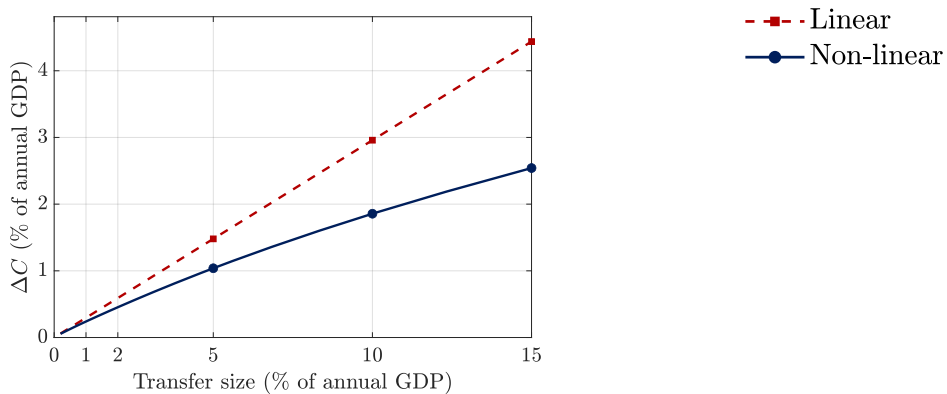
(b) MPC distribution



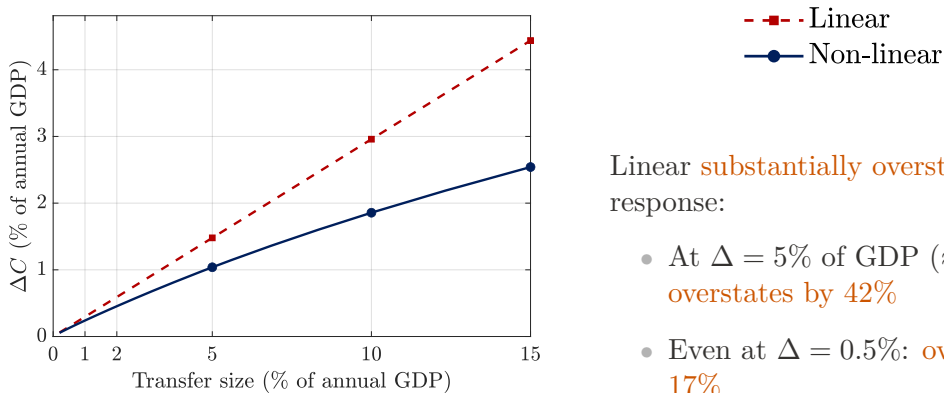
Aggregate consumption response



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Linear **substantially overstates** response:

- At $\Delta = 5\%$ of GDP ($\approx \$8,300$): **overstates by 42%**
- Even at $\Delta = 0.5\%$: **overstates by 17%**

Concavity and the Kink

- MPC is decreasing in wealth and **drops discontinuously** at the kink

$$m(x, y) = \begin{cases} 1, & x < \bar{x}(y), \\ m^+ & x > \bar{x}(y). \end{cases}$$

$$\frac{\beta \mathbb{E}[V_{aa}(a'(x, y), y') \mid y]}{u''(c) + \beta \mathbb{E}[V_{aa}(a'(x, y), y') \mid y]}$$

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- Non-linear: transfer may move constrained households **off the constraint**: their MPC drops from 1 to m^+

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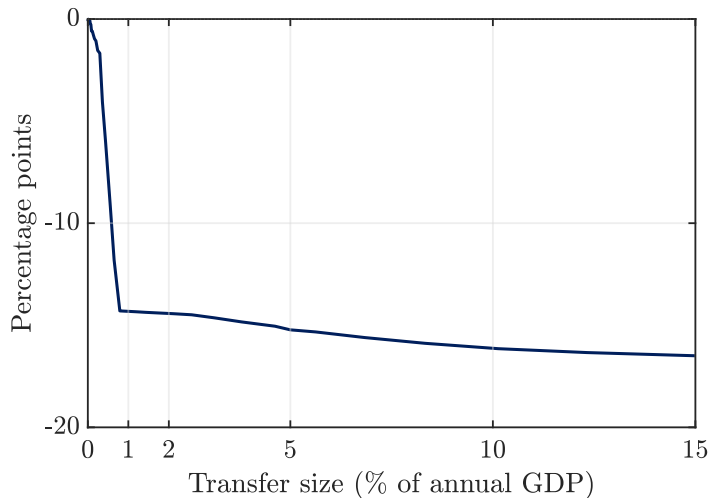
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- Aggregate MPC $\downarrow$ as $\Delta \uparrow \Rightarrow$ **concave** aggregate consumption response

Change in share of constrained households



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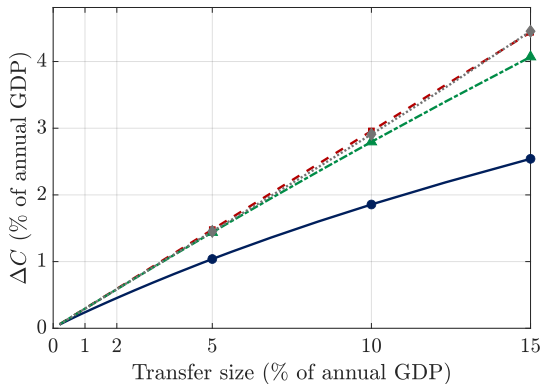
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- Unconstrained households: 2nd and 3rd order Taylor expansion

Analytical Derivatives

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Analytical Derivatives



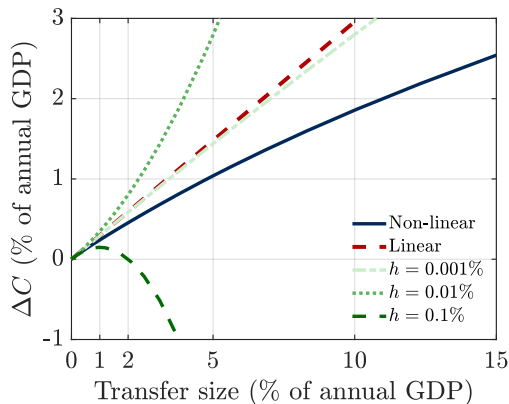
- ■ --- Linear
- ● — Non-linear
- - ▲ - - Second-order
- ... ◆ ... Third-order

- Second-order: little improvement
- Third-order: worse for shocks $> 5\%$

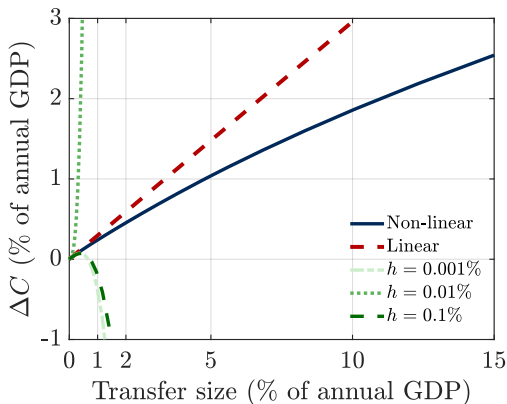
Constrained vs. unconstrained →

Higher-order with finite differences

(a) Second order



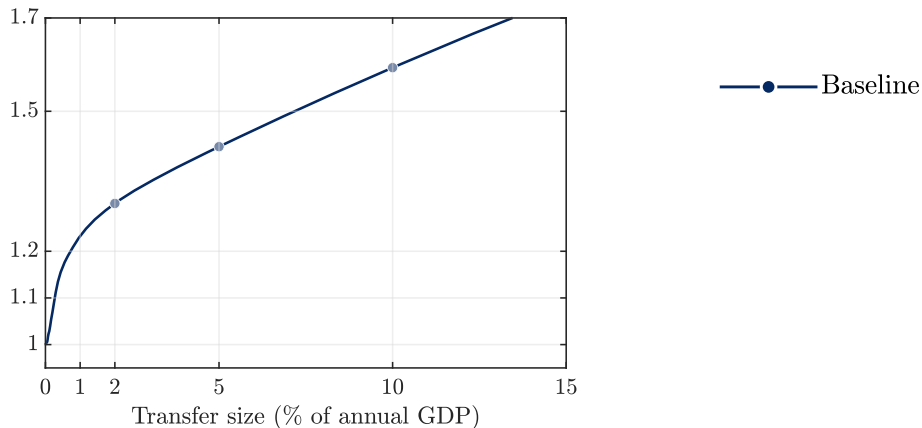
(b) Third order



Extremely sensitive to step size and poorly behaved even for small shocks.

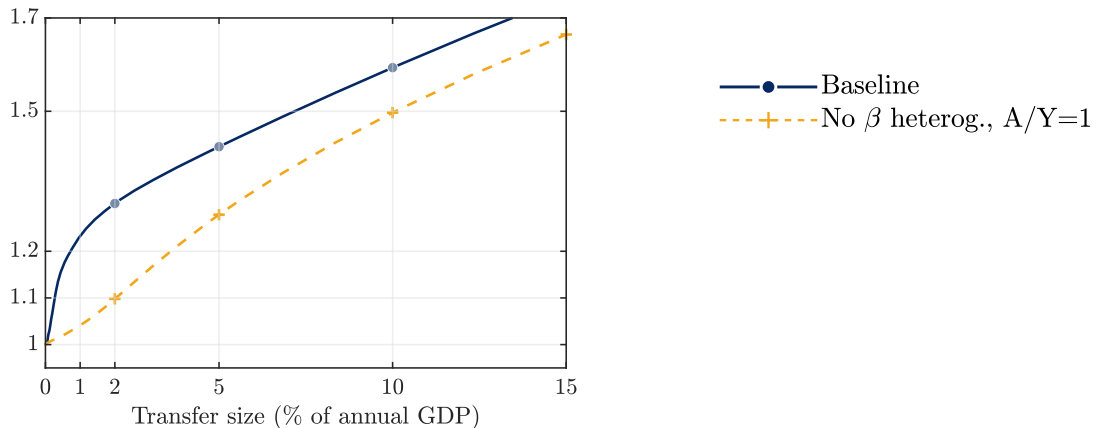
Robustness of Partial Equilibrium Results

Linear/Non-linear Consumption Response



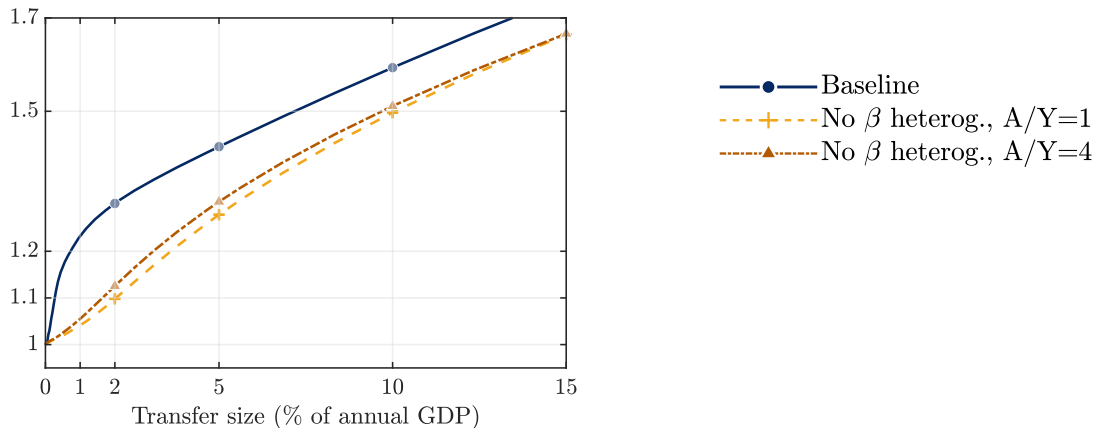
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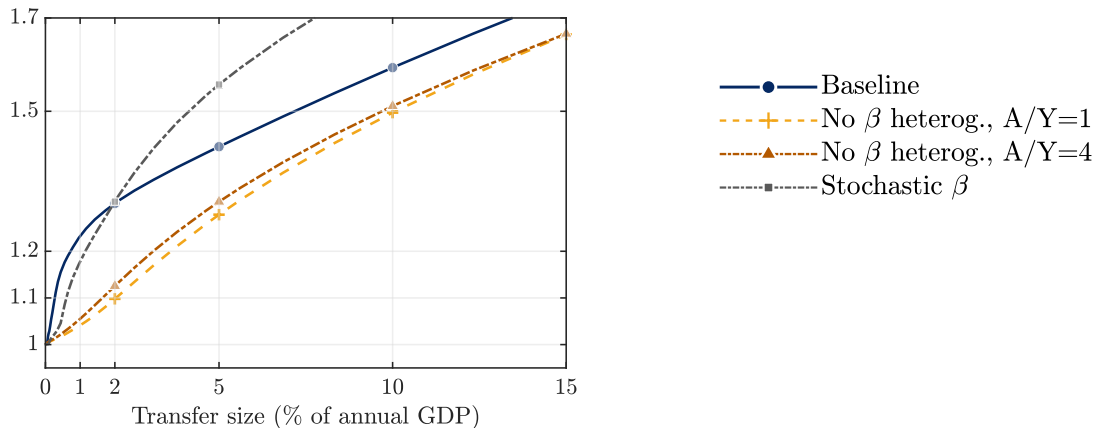
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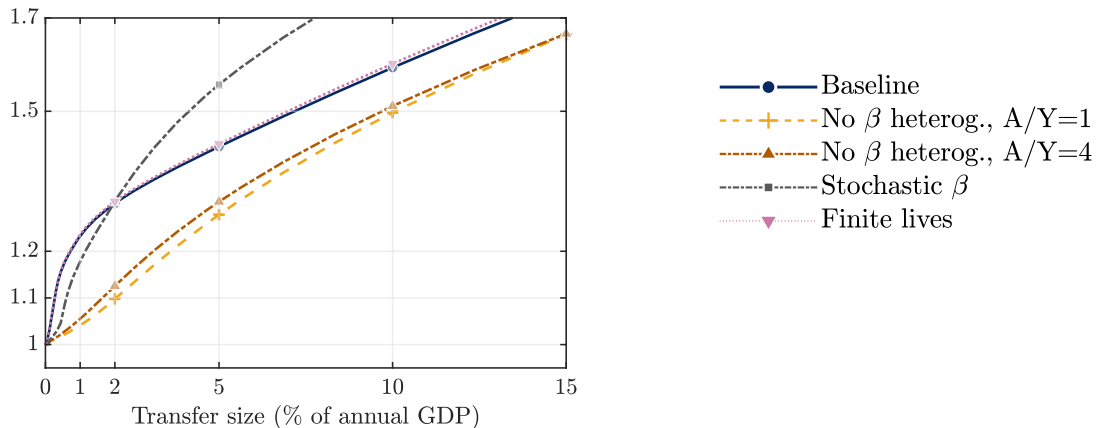
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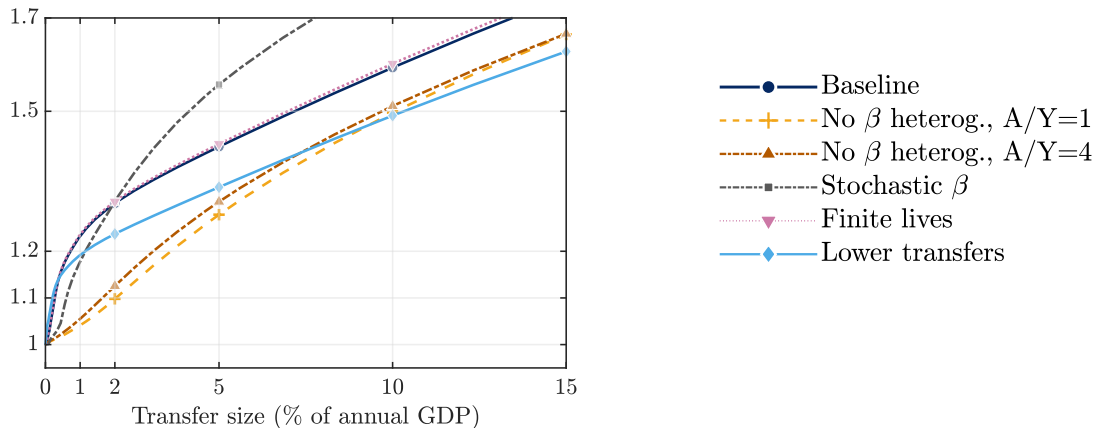
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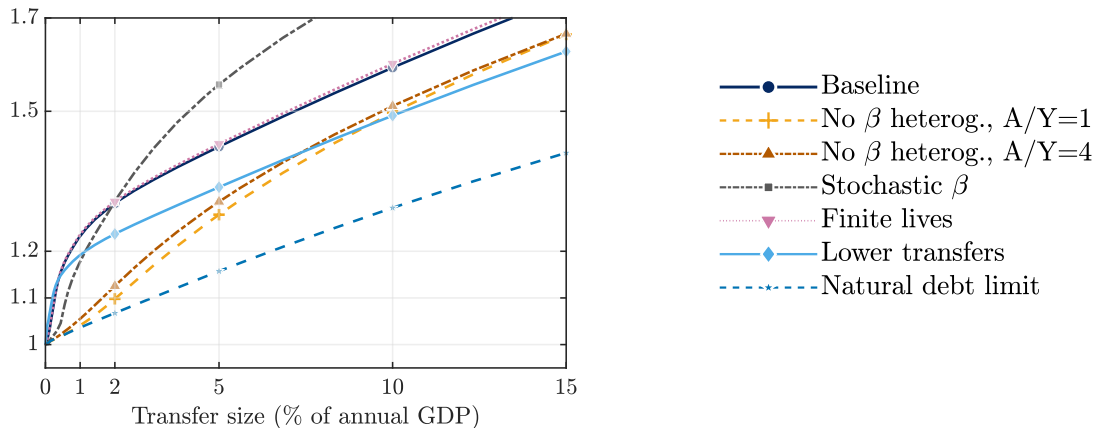
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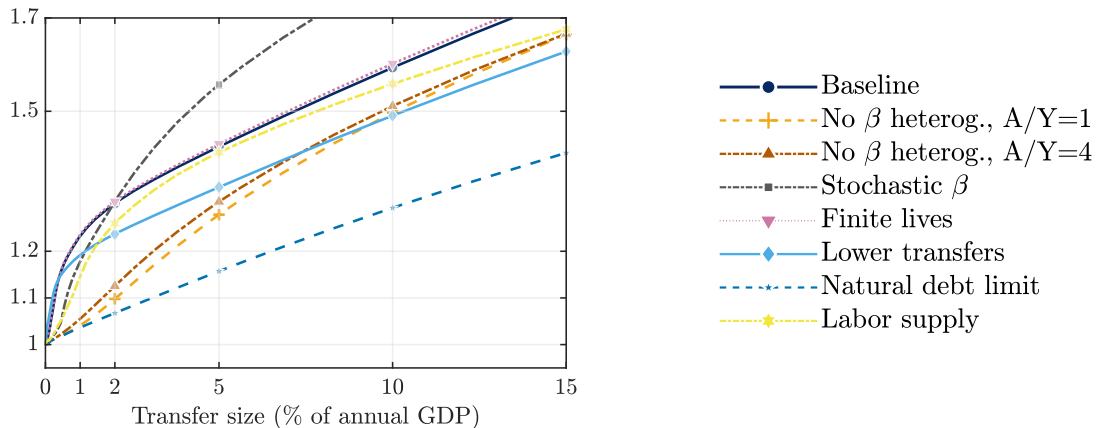
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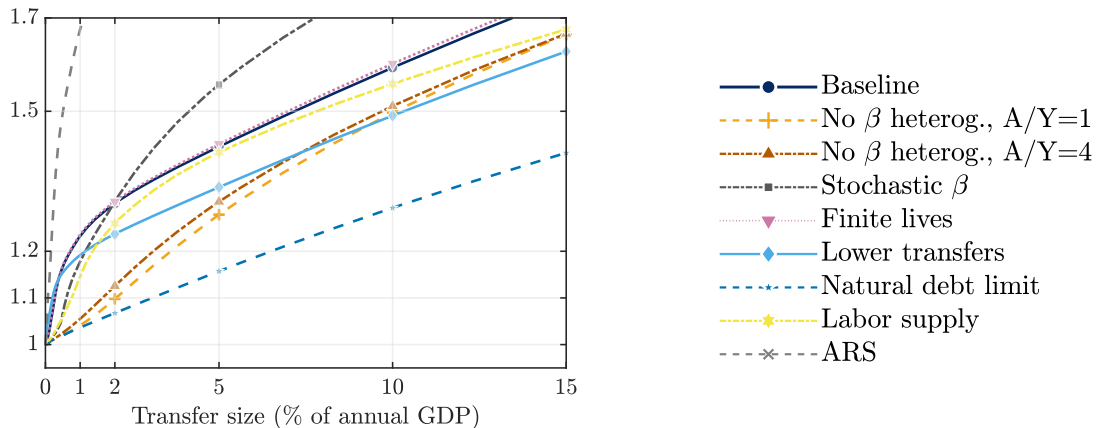
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Roadmap

1. Partial equilibrium model and calibration
2. Fiscal stimulus in general equilibrium
3. Other shocks

General Equilibrium Model

PE + production, labor supply, nominal rigidities, fiscal/monetary rules.

General Equilibrium Model

- **Households:**

- ▶ Utility flow, $u(c_{it}) - v(n_{it})$ with $v' > 0, v'' > 0$
- ▶ Income $(1 - \tau) \left(\frac{W_t}{P_t} e_{it} n_{it} + d_{i,t} \right)$

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- ▶ Labor union with sticky wages (linearized)

$$\pi_t = \kappa \left(v'(N_t) - \frac{1 - \tau}{C_t} \right) + \bar{\beta} \pi_{t+1}, \quad (\text{Phillips Curve})$$

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- **Firms and Technology:**

$$\left. \begin{array}{l} \text{Production: } Y_t = N_t \\ \text{Flexible prices} \end{array} \right\} \Rightarrow W_t = P_t, \quad d_{i,t} = 0$$

General Equilibrium Model

Government Policies

General Equilibrium Model

Government Policies

- Government budget:

$$B_t + \tau Y_t = G + T_t + (1 + r_{t-1})B_{t-1}$$

- Fiscal rule ($t > 0$)

$$T_t = T - \phi_B \bar{r} (B_{t-1} - \bar{B})$$

- Monetary policy:

$$i_t = \bar{i} + \phi_\pi (\pi_t - \bar{\pi})$$

Calibration

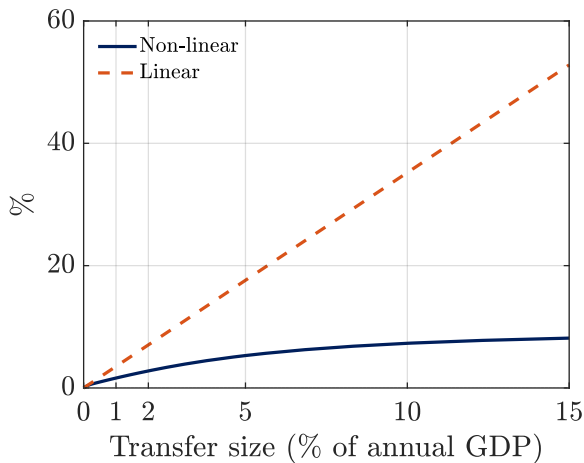
Parameter	Value	Description
$\bar{B}$	4	Steady-state government debt=100% GDP
$\bar{r}$	0.02	Steady-state annual real interest rate
$\bar{\pi}$	0	Steady-state inflation
κ	0.01	Phillips curve slope
ν	1	Frisch elasticity of labor supply
ϕ_{π}	1.5	Taylor coefficient
ϕ_B	8	Fiscal rule
G	0.104	Government spending
T	0.15	Government Transfers

Computation:

- Linear: sequence space Jacobian (ABRS)
- Non-linear: quasi-Newton algorithm

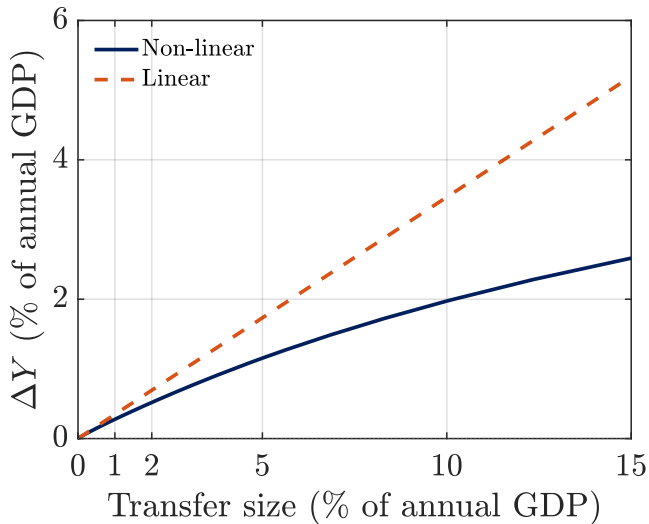
Flexible-Price Allocation: Real Interest Rate

Real rate response



- No output effects
- Linear methods overstate increase in demand and rise in r

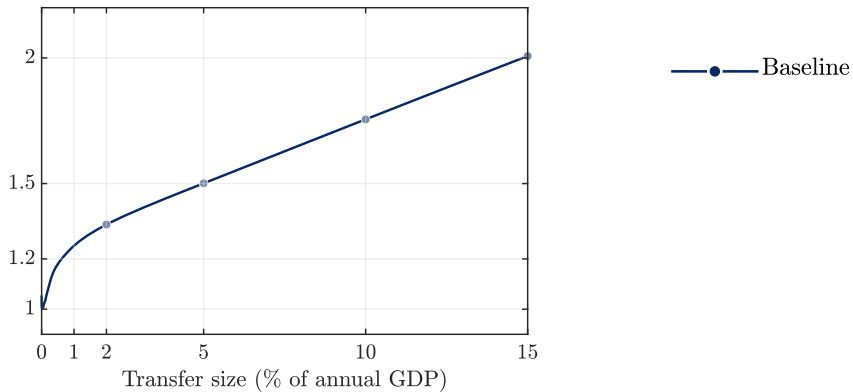
HANK: Output Response



- At $\Delta = 2\%$ GDP: linear overstates by 34%
- At $\Delta = 5\%$ GDP: linear overstates by 50%

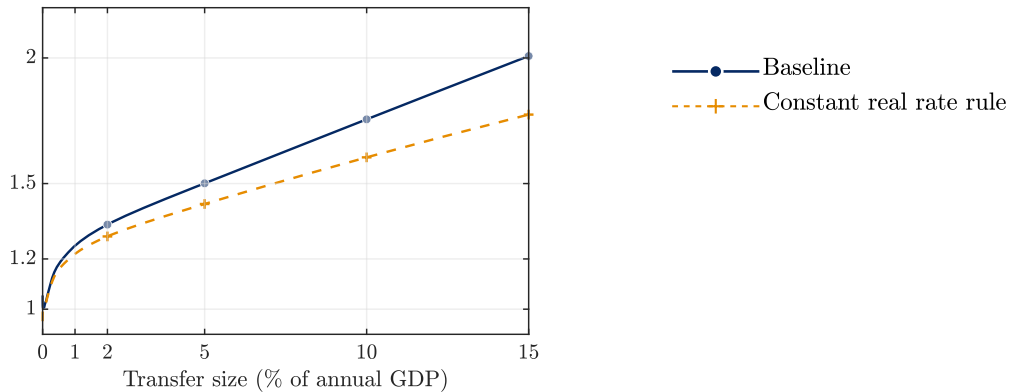
Robustness in HANK

Linear/Non-linear Output Response



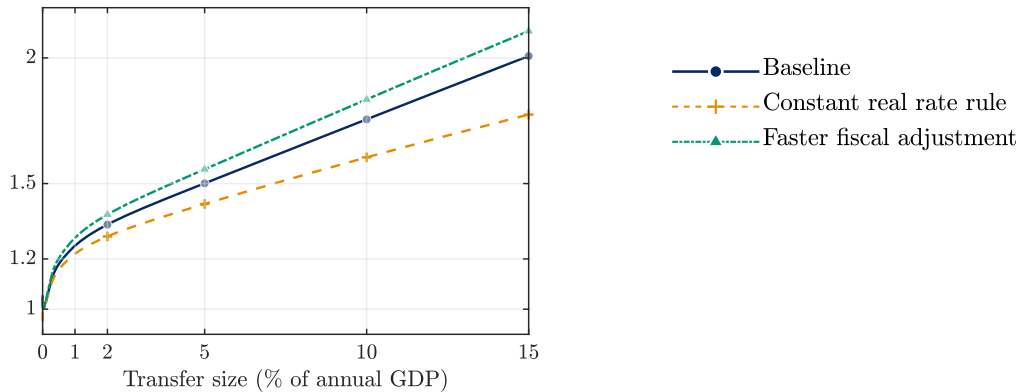
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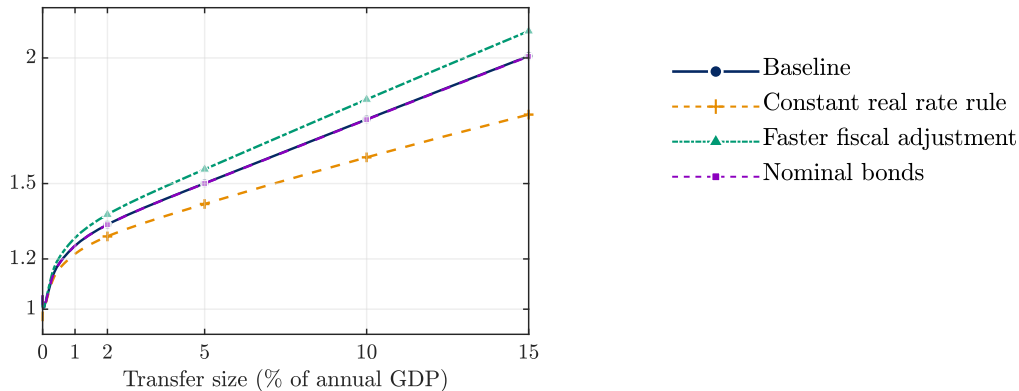
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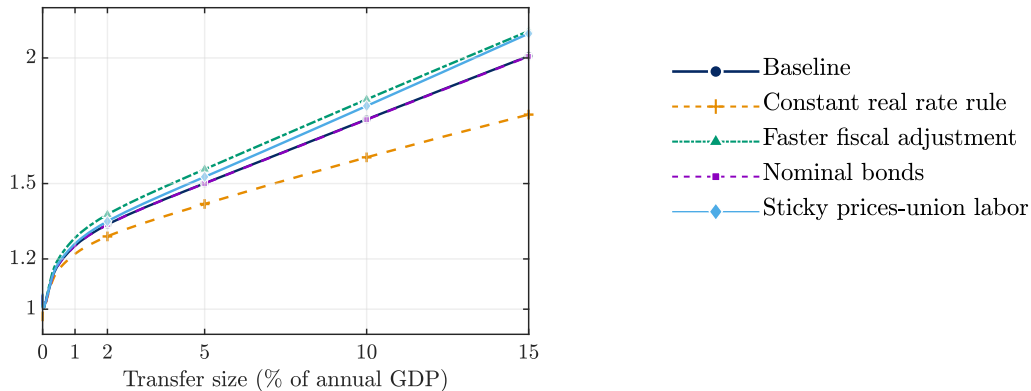
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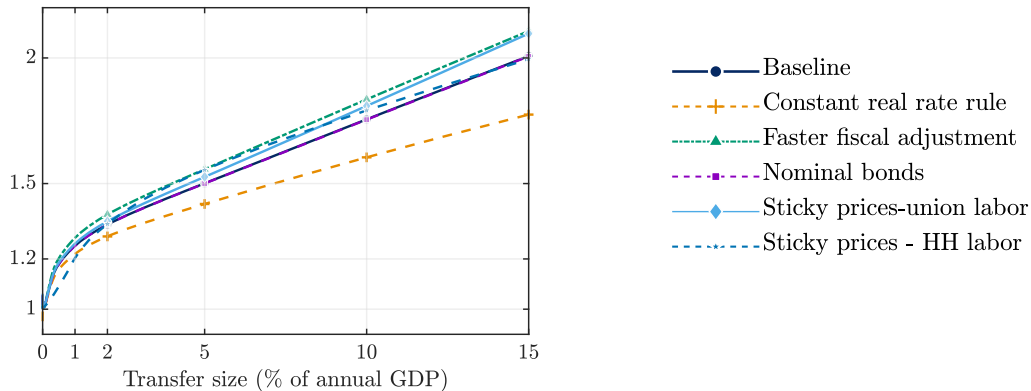
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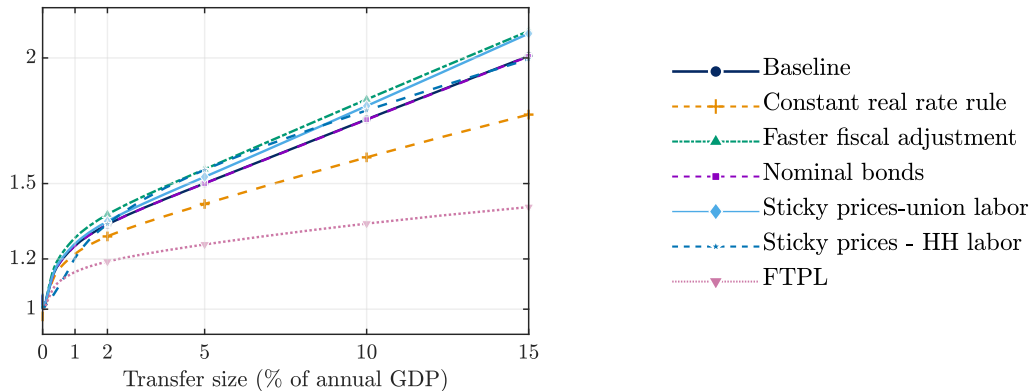
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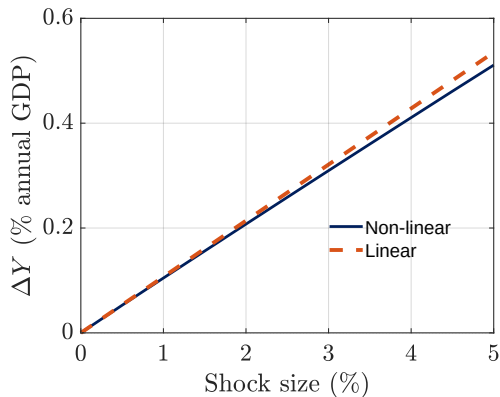
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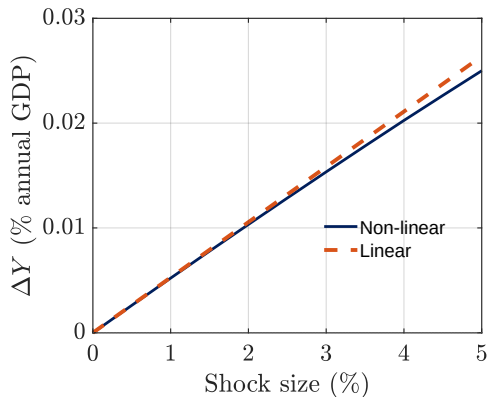


Non-linearities for monetary policy and TFP

(a) Monetary (rate cut)



(b) TFP



Conclusions

- HA models are valuable for analyzing fiscal and monetary policy because they capture realistic household behavior
- For monetary policy, linear approximation is accurate. But $HA \approx RA$
- For fiscal stimulus: need non-linear methods
 - ▶ Works through departures from Ricardian equivalence, where MPCs are **strongly non-linear in wealth**:
 - ▶ First-order methods (SSJ) treat constrained households as **permanently $MPC = 1$** , overstating their true response. **Higher-order local methods do not fix this**

For local methods to be reliable for fiscal stimulus, “small” must be *very* small.

Appendix

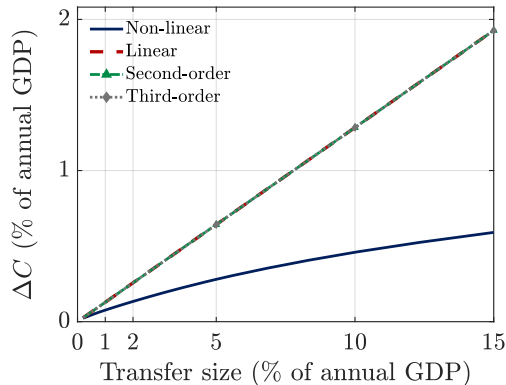
Additional figures

Implied parameters

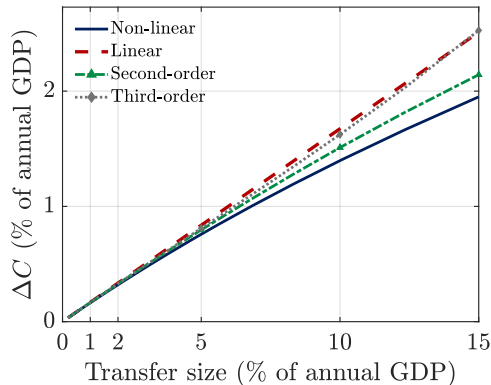
Parameter	Value
<i>Preferences & assets</i>	
r (annual)	2%
$\bar{\beta}$ (mean disc. factor)	0.972
σ_{β} (dispersion)	0.010
<i>Stochastic income process (Kaplan–Violante 2022)</i>	
ϕ_z (AR persistence)	0.988
σ_{η}^2 (perm. var.)	0.044
σ_{ε}^2 (trans. var.)	0.638
λ (shock arrival)	1/4 per qtr

Constrained vs. Unconstrained

Initially constrained



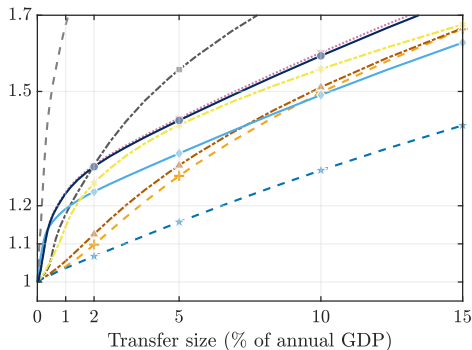
Initially unconstrained



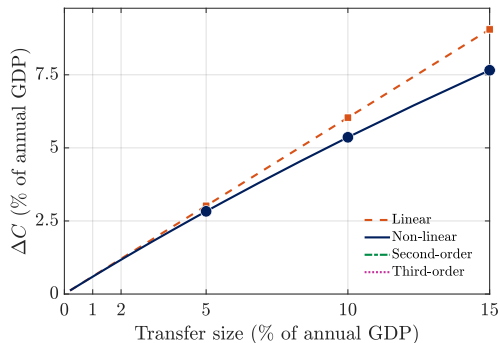
Note: Consumption response within each group, as a % of annual GDP per capita.

PE: KMV calibration and first-year response

(a) KMV calibration



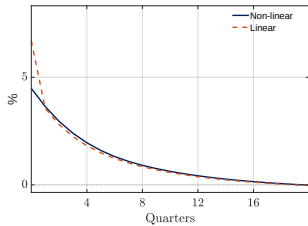
(b) First-year response



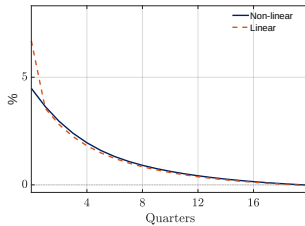
Note: Concavity is robust to the (Kaplan, Moll & Violante 2018) calibration and remains over the first year (panel b), though smaller than on impact.

GE baseline: impulse responses

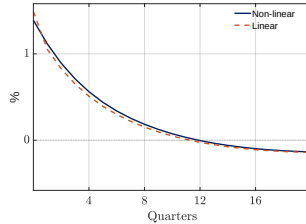
Output



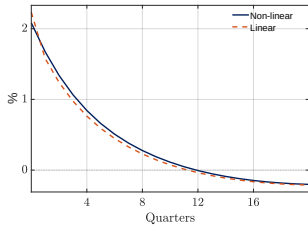
Consumption



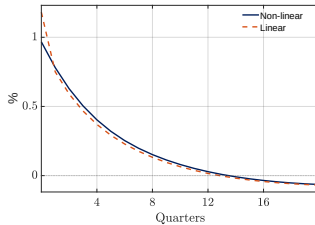
Inflation



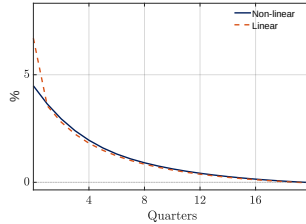
Nominal rate



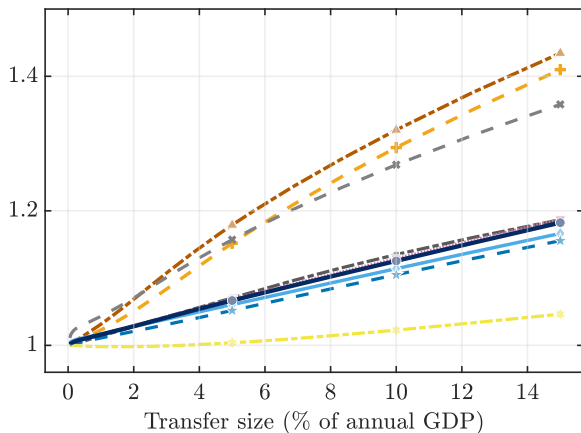
Real rate



Labor

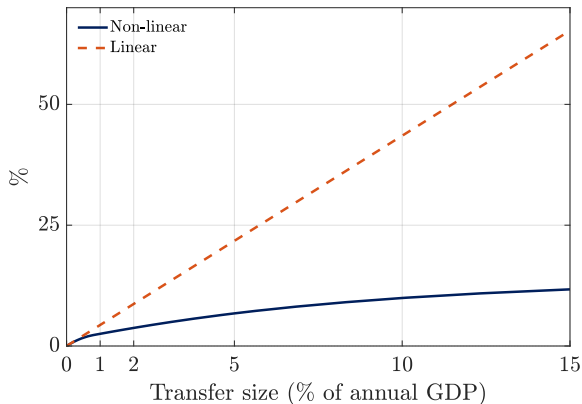


GE: first-year linear-to-non-linear ratio



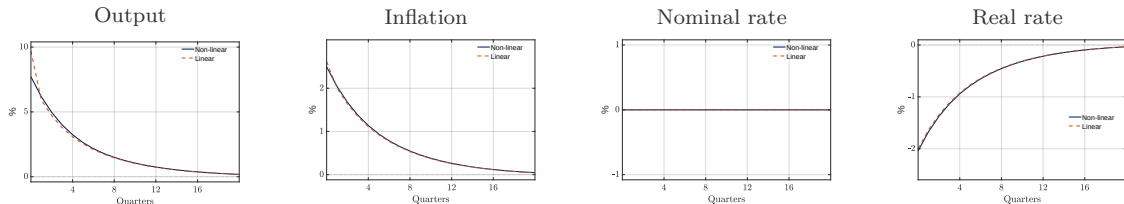
Note: Over the first year the linear method still overstates the output response, though by less than on impact.

GE: flexible prices with household labor choice



- Households choose labor freely (no union)
- Small output effects via labor-supply incentives
- Real-rate response remains **concave** in Δ — same mechanism as the union case

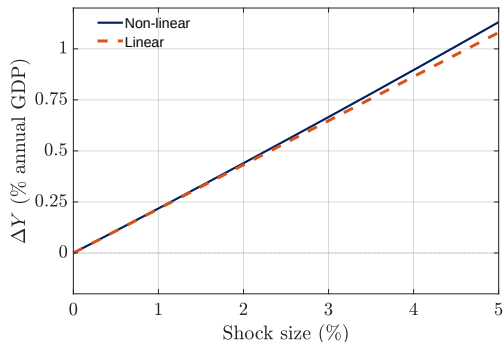
GE: unfunded stimulus (FTPL) impulse responses



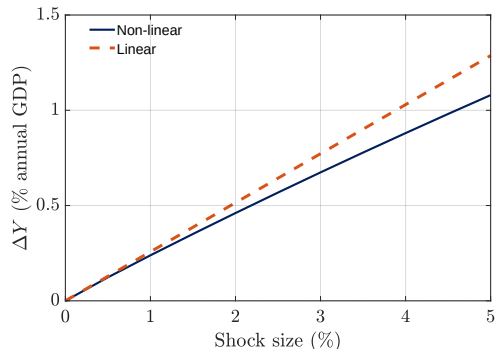
Note: Under the FTPL (active fiscal, passive money), the output effect does not hinge on a failure of Ricardian equivalence, so non-linearity is much milder.

Other shocks: discount factor and labor tax

(a) Discount-factor shock



(b) Labor-tax cut



Note: Both generate **milder non-linearity** than lump-sum transfers; a τ cut shifts fewer resources to low-income, constrained households.